
LATENTGAUGE: Prediction-Preserving Latent Geometry Attacks on World-Model Planning

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Abstract

World models increasingly plan by rolling actions forward in a learned latent space and scoring predicted states with Euclidean distance to a goal. LeWorldModel (LeWM) makes this geometry unusually explicit: it combines next-latent prediction with an isotropic-Gaussian anti-collapse regularizer, while downstream planning minimizes latent distance. Recent straightening and identifiability analyses suggest a benign picture in which learned coordinates become locally simple and, under ideal assumptions, are identifiable up to an orthogonal map. We expose the corresponding attack boundary. Gaussian marginal matching does *not* certify Euclidean planning geometry: a nonlinear transformation can preserve the standard-Gaussian latent distribution while changing distances, angles, temporal curvature, rollout conditioning, and the ranking of candidate action sequences. We introduce LATENTGAUGE, a planner-side attack that leaves the frozen encoder and predictor untouched and inserts only a Gaussian-measure-preserving remapping before the planner scores rollouts. Locally, the attack replaces the Euclidean metric by $G_T = DT^\top DT$, turning the planning Gauss-Newton matrix into $2J^\top G_T J$. We derive an exact feasible family from Gaussian-divergence-free flows, including radius-dependent rotations with a closed-form map, Jacobian, and induced metric, and optimize them for targeted trajectory-ranking flips under a latent-displacement budget. The resulting threat model is deliberately sharp: prediction can remain unchanged and Gaussian statistics can remain nominal while planning fails because the *geometry read by the planner* has been hijacked.

1 Problem setting: the geometry trusted by a world-model planner

LeWM learns an encoder h_θ and an action-conditioned predictor F_ϕ from pixel trajectories,

$$z_t = h_\theta(o_t), \quad \hat{z}_{t+1} = F_\phi(z_t, a_t), \quad \mathcal{L}_{\text{LeWM}} = \underbrace{\|\hat{z}_{t+1} - z_{t+1}\|_2^2}_{\mathcal{L}_{\text{pred}}} + \lambda \text{SIGReg}(Z), \quad (1)$$

where SIGReg pushes the aggregate latent distribution toward $\gamma = \mathcal{N}(0, I_d)$ and prevents collapse [1]. Given an initial latent z_0 , a horizon- H action sequence $\mathbf{a} = (a_0, \dots, a_{H-1})$, and a goal latent $z_g = h_\theta(o_g)$, define the recursive rollout

$$z_0^{\mathbf{a}} = z_0, \quad z_{t+1}^{\mathbf{a}} = F_\phi(z_t^{\mathbf{a}}, a_t), \quad \Phi_H(z_0, \mathbf{a}) := z_H^{\mathbf{a}}. \quad (2)$$

A standard terminal latent planner chooses

$$\mathbf{a}^* = \arg \min_{\mathbf{a}} C(\mathbf{a}), \quad C(\mathbf{a}) = \|\Phi_H(z_0, \mathbf{a}) - z_g\|_2^2. \quad (3)$$

Thus training constrains *prediction* and an aggregate *distribution*, while planning additionally trusts a particular *metric*: Euclidean distance in the learned coordinates.

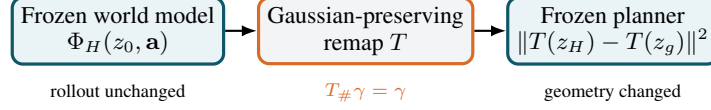


Figure 1: LATENTGAUGEattacks the latent-to-planner interface, not the predictive dynamics.

The text appendix derives why temporal straightening is useful for this planner and why, under strong identifiability assumptions, the remaining coordinate freedom can reduce to a common orthogonal transformation [2, 3, 4]. An orthogonal map Q is harmless because

$$Q^\top Q = I \implies \|Qx - Qy\|_2 = \|x - y\|_2, \quad (4)$$

so trajectory rankings, angles, and local least-squares conditioning are preserved.

Attack boundary. The security question is the converse: *can a latent interface look perfectly Gaussian while no longer being Euclidean-isometric?* If so, a deployment-time component can leave the world model statistically nominal yet alter the geometry used to choose actions. LATENTGAUGE targets exactly this gap.

2 Threat model and the induced planning metric

Planner-side threat model. The encoder h_θ , predictor F_ϕ , observations, actions, and goal image are frozen. The attacker can insert or corrupt a deterministic differentiable remapping $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ only at the interface where the planner scores predicted terminal states and goals. The clean rollout eq. (2) is unchanged; the planner instead evaluates

$$C_T(\mathbf{a}) = \|T(\Phi_H(z_0, \mathbf{a})) - T(z_g)\|_2^2. \quad (5)$$

We impose two stealth constraints,

$$T_{\#}\gamma = \gamma, \quad \gamma = \mathcal{N}(0, I_d), \quad D(T) := \mathbb{E}_{z \sim \gamma} \|T(z) - z\|_2^2 \leq \varepsilon^2. \quad (6)$$

The first preserves the target latent marginal exactly; the second limits average displacement from the clean coordinate system. Since T is applied *after* prediction, the raw one-step prediction residuals $\hat{z}_{t+1} - z_{t+1}$ are exactly unchanged.

Proposition 1 (Prediction and marginal stealth do not imply planning invariance). *Assume eq. (6). The frozen world model produces exactly the same clean latent rollouts and the planner-side transformed latents still have marginal distribution $\mathcal{N}(0, I_d)$. Nevertheless, if*

$$G_T(z) := DT(z)^\top DT(z) \neq I \quad (7)$$

on a region visited by planning rollouts, Euclidean planning costs are generally changed.

To see the mechanism, write $z = z_g + \delta$ and expand around z_g :

$$T(z_g + \delta) = T(z_g) + DT(z_g)\delta + \frac{1}{2}D^2T(z_g)[\delta, \delta] + O(\|\delta\|^3), \quad (8)$$

$$\|T(z_g + \delta) - T(z_g)\|_2^2 = \delta^\top G_T(z_g)\delta + O(\|\delta\|_2^3). \quad (9)$$

Thus T replaces the planner’s identity metric by the pullback metric G_T . Orthogonal maps satisfy $G_T \equiv I$; nonlinear Gaussian-preserving maps need not.

Effect on action optimization. Let

$$r_T(\mathbf{a}) = T(\Phi_H(z_0, \mathbf{a})) - T(z_g), \quad J(\mathbf{a}) = \frac{\partial \Phi_H(z_0, \mathbf{a})}{\partial \mathbf{a}}. \quad (10)$$

The residual Jacobian is

$$\tilde{J}_T(\mathbf{a}) = DT(\Phi_H(z_0, \mathbf{a})) J(\mathbf{a}). \quad (11)$$

Hence

$$\nabla_{\mathbf{a}} C_T = 2J^\top DT(\Phi_H)^\top r_T, \quad (12)$$

$$H_T^{\text{GN}} = 2\tilde{J}_T^\top \tilde{J}_T = 2J^\top G_T(\Phi_H)J. \quad (13)$$

The same G_T therefore changes both *which rollout is cheaper* and *how well conditioned gradient-based planning is*. In particular, the relevant nonzero spectrum becomes that of $G_T^{1/2} J$, rather than J alone.

Effect on temporal straightness. For a continuous clean latent curve $z(s)$, the transformed curve $\tilde{z}(s) = T(z(s))$ obeys

$$\dot{\tilde{z}}(s) = DT(z(s))\dot{z}(s), \quad (14)$$

$$\ddot{\tilde{z}}(s) = D^2T(z(s))[\dot{z}(s), \dot{z}(s)] + DT(z(s))\ddot{z}(s). \quad (15)$$

Even if the clean trajectory is locally straight, $\ddot{z} = 0$, a nonlinear T can create curvature through the Hessian term $D^2T[\dot{z}, \dot{z}]$. Thus Gaussianity alone cannot certify the straight geometry that motivates latent planning.

3 Exact Gaussian-preserving attacks by divergence-free flow

A generic neural warp is easy to optimize but difficult to keep exactly distribution-preserving. We instead parameterize T as the time- τ flow of a smooth vector field v_ψ :

$$\frac{dz(s)}{ds} = v_\psi(z(s)), \quad z(0) = z, \quad T_\psi(z) := z(\tau). \quad (16)$$

Let $\rho(z) = (2\pi)^{-d/2} \exp(-\|z\|^2/2)$ be the standard-Gaussian density. Density evolution under the flow follows the continuity equation

$$\partial_s \rho_s + \text{div}(\rho_s v_\psi) = 0. \quad (17)$$

Therefore $\rho_s \equiv \rho$ is stationary, and the flow preserves γ , whenever

$$\text{div}(\rho v_\psi) = 0 \iff \text{div } v_\psi(z) + \nabla \log \rho(z)^\top v_\psi(z) = 0 \iff \boxed{\text{div } v_\psi(z) - z^\top v_\psi(z) = 0}. \quad (18)$$

This converts a global distribution constraint into a structural local constraint on the attack field.

Proposition 2 (Radius-dependent rotational family). *Let $A^\top = -A$ be any skew-symmetric matrix and let $\alpha_\psi : \mathbb{R}_+ \rightarrow \mathbb{R}$ be any smooth scalar function. Then*

$$v_\psi(z) = \alpha_\psi(\|z\|_2^2) Az \quad (19)$$

satisfies eq. (18). The flow preserves $\|z\|_2$ and the standard-Gaussian measure exactly.

Indeed, using $z^\top Az = 0$ and $\text{tr}(A) = 0$,

$$z^\top v_\psi(z) = \alpha_\psi(\|z\|_2^2) z^\top Az = 0, \quad (20)$$

$$\text{div } v_\psi(z) = 2\alpha'_\psi(\|z\|_2^2) z^\top Az + \alpha_\psi(\|z\|_2^2) \text{tr}(A) = 0, \quad (21)$$

$$\frac{d}{ds} \|z(s)\|_2^2 = 2z(s)^\top v_\psi(z(s)) = 0. \quad (22)$$

Because the radius $r^2 = \|z\|_2^2$ is constant along the flow, eq. (16) has the closed form

$$\boxed{T_\psi(z) = \exp(\tau \alpha_\psi(\|z\|_2^2) A) z}. \quad (23)$$

At each fixed radius the exponential is orthogonal, so the map keeps the Gaussian shell structure exactly. If α_ψ is constant, eq. (23) reduces to one global orthogonal rotation and cannot attack Euclidean planning. If $\alpha'_\psi \neq 0$, different radii rotate by different amounts, producing a nonlinear, generally non-isometric warp.

Closed-form local metric distortion. Let $s = \|z\|_2^2$, $R(s) = \exp(\tau \alpha_\psi(s) A)$, and $\beta(s) = \tau \alpha'_\psi(s)$. Since A commutes with $R(s)$,

$$DT_\psi(z) = R(s) [I + 2\beta(s) A z z^\top]. \quad (24)$$

As $R(s)^\top R(s) = I$, the induced metric is

$$\begin{aligned} G_T(z) &= [I + 2\beta A z z^\top]^\top [I + 2\beta A z z^\top] \\ &= I + 2\beta (A z z^\top - z z^\top A) + 4\beta^2 (z^\top A^\top A z) z z^\top. \end{aligned} \quad (25)$$

This equation makes the attack visible: the *distribution* is unchanged, but the planner reads a position-dependent metric whenever $\beta \neq 0$.

For a general small feasible flow $T_\epsilon(z) = z + \epsilon v(z) + O(\epsilon^2)$,

$$G_{T_\epsilon}(z) = I + \epsilon (Dv(z) + Dv(z)^\top) + O(\epsilon^2). \quad (26)$$

The symmetric part of Dv is therefore the first-order object that the attacker uses to stretch or compress planner-sensitive directions.

4 Targeted attack objective

Trajectory-ranking flip. For an instance i , let $\mathbf{a}_i^+$ denote the clean preferred/safe candidate and $\mathbf{a}_i^-$ a targeted alternative. Define the clean margin

$$m_i^{\text{clean}} := C(\mathbf{a}_i^-) - C(\mathbf{a}_i^+) > 0. \quad (27)$$

Under the attack, define

$$m_i^T := C_T(\mathbf{a}_i^-) - C_T(\mathbf{a}_i^+). \quad (28)$$

A ranking flip occurs exactly when $m_i^T < 0$. A smooth targeted training loss with desired attack margin $\kappa \geq 0$ is

$$\mathcal{L}_{\text{rank}}(\psi) = \frac{1}{N} \sum_{i=1}^N \text{softplus}(m_i^T + \kappa), \quad (29)$$

which is minimized when $C_T(\mathbf{a}_i^-) + \kappa < C_T(\mathbf{a}_i^+)$. Because Gaussian preservation is structural, no SIGReg-matching penalty is needed. We enforce the displacement budget with

$$\mathcal{L}_{\text{atk}}(\psi) = \mathcal{L}_{\text{rank}}(\psi) + \lambda_D [D(T_\psi) - \varepsilon^2]_+^2. \quad (30)$$

Optional conditioning objective. Ranking reversal is the primary attack. A second, complementary objective targets optimization difficulty rather than a specific alternative trajectory:

$$\mathcal{L}_{\text{cond}}(\psi) = -\frac{1}{N} \sum_{i=1}^N \log \text{cond}_+(J_i^\top G_T(\Phi_i) J_i), \quad (31)$$

where cond_+ uses only nonzero singular/eigen directions. This directly attacks the Gauss–Newton geometry implicated by temporal straightening while leaving the dynamics fixed.

Optimization loop. We parameterize A by a small basis of skew matrices A_k and $\alpha_\psi(s)$ by a low-dimensional radial basis, for example

$$A(\eta) = \sum_{k=1}^{K_A} \eta_k A_k, \quad \alpha_\psi(s) = \sum_{j=1}^{K_r} c_j \varphi_j(s). \quad (32)$$

For each mini-batch: (1) roll out the frozen world model for candidate action sequences; (2) compute terminal latents and the goal latent; (3) apply the closed-form T_ψ from eq. (23); (4) compute eq. (30) (and optionally eq. (31)); and (5) backpropagate only to $\psi = (\eta, c)$. The encoder, predictor, candidate rollouts, and target goal encoding are never updated.

LATENTGAUGE in one sentence. Keep the world model’s predictions fixed, move only within an exactly Gaussian-preserving family of latent coordinate systems, and choose the nonlinear coordinate warp that makes the planner prefer the wrong rollout.

5 Evaluation: separate prediction, statistics, and planning

The empirical question is not whether an arbitrary latent warp can break a planner; it can. The test is whether *exactly Gaussian-preserving, small-displacement* gauge freedom is sufficient to produce meaningful planning failures while all prediction-side diagnostics remain clean.

Latent map	Gaussian	Prediction	Euclidean geometry	Attack
Identity	preserved	unchanged	preserved	none
Orthogonal Q	preserved	unchanged	preserved	negative control
Random radial twist	preserved	unchanged	changed	untargeted
LATENTGAUGE	preserved	unchanged	changed	targeted
Unconstrained MLP	not guaranteed	unchanged	changed	detectable upper bound

Table 1: The key control is orthogonal gauge invariance: only nonlinear Gaussian-preserving maps can remain marginally nominal while changing Euclidean planning geometry.

Models and tasks. We propose two levels. First, use a controlled synthetic dynamical system with known state geometry and known optimal action rankings; this directly validates eq. (25). Second, evaluate a LeWM-style pixel world model on goal-conditioned latent planning, preserving its original encoder, predictor, and candidate-generation procedure.

Baselines. Use Identity; random orthogonal Q ; optimized orthogonal Q ; random radial twist; optimized LATENTGAUGE; and an unconstrained MLP warp as a non-stealth upper bound. The orthogonal baselines are essential: if they change the planner materially, the implementation itself violates the intended Euclidean invariance.

Metrics. Report the three axes separately:

$$\text{prediction: } \text{MSE}_{\text{pred}} = \mathbb{E} \|F_\phi(z_t, a_t) - z_{t+1}\|_2^2, \quad (33)$$

$$\text{statistics: } D(T) = \mathbb{E} \|T(z) - z\|^2, \quad \text{SIGReg}(T(Z)), \quad (34)$$

$$\text{planning: } \text{FlipRate} = \frac{1}{N} \sum_i \mathbf{1}[m_i^T < 0], \quad (35)$$

$$\Delta C_{\text{clean}} = \mathbb{E}_i [C(\hat{\mathbf{a}}_{T,i}) - C(\hat{\mathbf{a}}_{\text{clean},i})], \quad (36)$$

$$\Delta \mathcal{S}_{\text{str},i} = \mathcal{S}_{\text{str}}(T(z_{0:H}^{(i)})) - \mathcal{S}_{\text{str}}(z_{0:H}^{(i)}), \quad \mathcal{S}_{\text{str}}(z_{0:H}) = \frac{1}{H-1} \sum_{t=0}^{H-2} \frac{v_t^\top v_{t+1}}{\|v_t\| \|v_{t+1}\|}, \quad (37)$$

where $\hat{\mathbf{a}}_{T,i}$ is the action sequence selected by the attacked planner but $C(\cdot)$ in eq. (36) is always recomputed in the clean latent coordinates (or, where available, in ground-truth task space). Also report the nonzero singular spectrum of $G_T^{1/2} J$ and $\text{cond}_+(J^\top G_T J)$.

Ablations. Vary displacement budget ε , horizon H , number of skew directions K_A , radial capacity K_r , and candidate-set size. Compare global versus instance-specific T . Measure transfer across unseen goals and trajectories to distinguish memorized ranking manipulation from a genuine geometric attack. Finally, stratify by latent radius because eq. (23) explicitly couples distortion strength to Gaussian shells.

Falsifiable claims. This extended abstract supports three mathematical claims and leaves the empirical magnitude open:

1. Gaussian marginal preservation plus unchanged prediction does not certify Euclidean planner invariance (proposition 1).
2. Planner ranking and Gauss–Newton conditioning change through the induced metric $G_T = DT^\top DT$ (eqs. (9) and (13)).
3. Nontrivial exactly Gaussian-preserving, nonlinear attacks exist constructively (eqs. (23) and (25)).

The experiments test only the remaining question: whether this mathematically available gauge freedom is large enough, under realistic budgets, to flip action rankings and degrade task success in finite LeWM models.

6 Implications, defenses, and scope

What the attack demonstrates. A low SIGReg value is a statement about the *distribution of points*; it is not a certificate of the *metric between points*. Likewise, low prediction error is a statement about the frozen transition model; in the planner-side threat model it says nothing about a downstream remapping. Consequently, the pair “prediction remains correct + latent cloud remains Gaussian” is insufficient evidence that the planner is operating in the geometry that training produced.

What would defend the interface. The attack suggests that a stronger deployment certificate must constrain geometry, not only marginals. Minimal candidates include checking local isometry,

$$\mathcal{E}_{\text{iso}} = \mathbb{E}_{z \sim \gamma} \|DT(z)^\top DT(z) - I\|_F^2, \quad (38)$$

checking anchored pairwise distances on trusted latent pairs, or cryptographically/integrity-protecting the latent-to-planner path so no untrusted remapping can be inserted. These are defenses implied by the threat model, not claims that one particular certificate is sufficient in all world models.

Scope. The planner-side attack assumes the adversary can alter the latent scoring interface but not the world-model rollout itself. That makes the separation unusually clean: any observed failure is geometric rather than predictive. It does not claim that LeWM is uniquely vulnerable, nor that every deployment exposes such an interface. The construction applies to any planner that trusts Euclidean distances in a latent space whose marginal distribution is used as a correctness or anti-collapse diagnostic.

Takeaway. The appendix asks why predictive learning may select a straight, planning-friendly latent gauge. LATENTGAUGE asks the complementary security question: *what if that gauge is changed after learning without changing the latent distribution?* Orthogonal ambiguity is safe; nonlinear Gaussian-preserving ambiguity is not. The core lesson is therefore concise:

$$\boxed{\text{correct prediction}} + \boxed{\text{Gaussian marginal}} \not\Rightarrow \boxed{\text{trusted planning geometry}}. \quad (39)$$

References

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